GENIE GROUNDING / KNOWLEDGE DOCUMENT

Helivara Supply Chain & Manufacturing

Reference context for the Genie conversational agents: portfolio, manufacturing network, governed metric views, and the demo's hero facts.

Helivara is a FICTIONAL pharmaceutical company. All data is synthetic and generated for demonstration purposes; the business processes and reasoning are industry-realistic. Product brand names are used only as illustrative examples.

Platform: Databricks (Unity Catalog, Genie, AI/BI, Discover)
Catalog / schema: fr_stable_catalog.supply_chain
Demo anchor date ("today"): 2026-09-22
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1. Purpose & how to use this document

This document is the shared knowledge base for Helivara's Genie agents. Attach it to a Genie space (or store it in a Unity Catalog Volume and reference it) so the assistant can ground its answers in consistent business context: what the products are, how the network is structured, which governed metrics to use, and which facts define the demo storyline.

The two Genie agents

- **Demand & Supply Agent:** S&OP, demand forecast accuracy and bias, launch readiness, order service (OTIF), inventory and expiry-at-risk.
- **Manufacturing & Reliability Agent:** OEE and its components, right-first-time and yield, quality deviations and CAPA, equipment reliability and predicted failures.

Answering rules for the assistant

- **Use the governed metric views:** prefer the mv_* views and wrap every metric in MEASURE(). Do not recompute metrics from raw fact tables when a metric view exists.
- **Cite the source:** name the metric view used in each answer.
- **Bilingual:** users may ask in Italian; answer in Italian when they do.
- **Always flag the launch at risk:** if a question touches launches or Zyvanta, surface that it is at risk and name the blocker (master data / artwork).

2. Company & manufacturing network

Helivara is a mid-size specialty + consumer-health pharmaceutical manufacturer. The demo models an end-to-end "plant-to-patient" flow: Demand & S&OP -> Supply & Inventory -> Manufacturing & Quality -> Reliability & Maintenance.

Plants (5)

Plant	Site type	Country	Role
Valmora	OSD	Italy	Primary oral-solid-dose hub; Zyvanta coating & packaging
Castelverde	API	Italy	Active ingredient (granulation / processing)
Roccalba	OSD	Italy	Oral-solid-dose compression
Riberal	OSD	Spain	Oral-solid-dose compression
Ashbury Ridge	OSD	USA	Oral-solid-dose compression & packaging

Markets & distribution

Markets: Italy, Spain, EU, US, RoW (regions: Europe, Americas, Global). Each market is served by a distribution centre (DC_Italy, DC_Spain, DC_EU, DC_US, DC_RoW).

3. Product portfolio (11 brands)

Four therapeutic clusters/BUs: Immunology, Gastroenterology (GI), Vascular, Consumer Health. Launch phase drives the storyline: one launch (Zyvanta), one phase-out (Oribex), the rest mature.

Brand	Therapeutic area	Cluster/BU	Phase	Notes
Zyvanta	Immunology	Immunology	LAUNCH	Hero: launch AT RISK (see sec. 5)
Oribex	Immunology	Immunology	phase-out	US sales = 0 after 2025-11
Gastrelle	Gastroenterology	GI	mature	
Colvera	Gastroenterology	GI	mature	

Brand	Therapeutic area	Cluster/BU	Phase	Notes
Florabis	Gastroenterology	GI	mature	Short shelf life (12 mo)
Lactavio	Gastroenterology	GI	mature	
Refluxa	Gastroenterology	GI	mature	
Venoria	Vascular	Vascular	mature	
Faringel	Consumer Health	Consumer	mature	Serialization exempt
Digestin	Consumer Health	Consumer	mature	Serialization exempt
Citralux	Consumer Health	Consumer	mature	Serialization exempt

4. Governed metric views

Eight Unity Catalog metric views define the certified business metrics. Query measures with MEASURE(<name>) and group by the listed dimensions. These are the assistant's preferred source.

Metric view	Key measures (use MEASURE())	Answers
mv_demand	forecast_accuracy, forecast_bias, consensus_vs_statistical_gap	Forecast accuracy & bias by therapeutic area, market, cluster/BU
mv_launch_readiness	avg_launch_readiness, master_data_ready_rate, artwork_ready_rate, serialization_ready_rate	Launch readiness and which gate is blocking
mv_service	otif, total_stockout_days, total_backorder_qty	On-time-in-full service by market / brand
mv_inventory	avg_days_of_inventory, total_on_hand, total_expiry_at_risk	Days of inventory and expiry-at-risk by DC / market
mv_manufacturing	oee, availability, performance, quality, total_downtime_min	OEE and its three components by plant / line / shift
mv_production	rft_rate, avg_yield, schedule_adherence, batch_count	Right-first-time, yield, schedule adherence by plant
mv_quality	deviation_count, major_count, critical_count, capa_on_time_rate	Deviations, severity and CAPA on-time by plant / root cause
mv_reliability	unplanned_downtime_hours, mttr_min, total_maintenance_cost, wo_count	Reliability, downtime and maintenance cost by equipment

Rule of thumb: OEE questions must report all three components (availability x performance x quality). Downtime cascades to supply: lower OEE -> schedule miss -> stockout -> lower OTIF (especially Zyvanta coating at Valmora).

5. Hero facts (the demo storyline)

Zyvanta (immunology) - launch AT RISK
Overall launch readiness: 0.33 (lowest in the portfolio).
Master data ready rate: 0% -> BLOCKER (material master incomplete).
Artwork ready rate: 0% -> BLOCKER (artwork still in review).
Serialization ready rate: 100% -> this gate is NOT the problem.
So: the launch is blocked by master data and artwork, NOT serialization.
Manufacturing link: Zyvanta is coated & packaged at Valmora; OEE loss there puts the launch supply at further risk.

Oribex (immunology) - phase-out
Being phased out; US sales = 0 after 2025-11.
Master data / artwork / serialization all complete (readiness ~0.9).
Do not treat declining Oribex demand as a forecasting error - it is planned.

Launch-readiness gates explained: a launch is 'ready' only when three governance gates pass - (1) material master data complete, (2) serialization status set, (3) artwork approved. Readiness is the average of the three, so a single failed gate materially lowers the score.

6. Discover domains

The catalog is organised into four governed domains, each with two subdomains, used in the Databricks Discover experience.

Domain	Subdomains
Demand and Sales Operations	Forecast; Consensus S&OP
Supply and Inventory	Production Planning; Inventory and Service
Manufacturing and Quality	Batch Execution and OEE; Quality and Deviations
Reliability and Maintenance	Asset Telemetry; Maintenance and Downtime

7. Glossary of key terms

- **S&OP:** Sales & Operations Planning - the monthly cross-functional process aligning demand, supply and finance on one plan.
- **Forecast bias:** Systematic over- or under-forecasting. Positive bias = over-forecast; near zero is healthy.
- **Forecast accuracy:** How close the forecast was to actual sales (higher is better).
- **Launch readiness:** Composite gate score (master data + serialization + artwork) that says whether a new product can launch.
- **Master data:** The governed product/material record (material master). Must be complete before launch.
- **Serialization:** Unique-code marking of packs for anti-counterfeiting / track-and-trace (some consumer packs are exempt).
- **Artwork:** Approved packaging/label content; must be signed off before launch.
- **OTIF:** On-Time-In-Full - orders delivered complete and on schedule.
- **Days of inventory (DIO):** How many days of demand current stock covers.
- **Expiry-at-risk:** Stock likely to expire before it can be sold.
- **OEE:** Overall Equipment Effectiveness = Availability x Performance x Quality.
- **RFT:** Right First Time - share of batches produced correctly without rework.
- **Deviation / CAPA:** A quality event and its Corrective And Preventive Action; on-time CAPA closure is a compliance KPI.
- **MTTR:** Mean Time To Repair - average time to restore failed equipment.
- **Unplanned downtime:** Equipment stoppage that was not scheduled - the costly kind.

8. Operational layer: predictions, value and actions

- **Failure risk (risk_score):** fact_predicted_failure is scored daily by the Unity Catalog model equipment_failure_model (alias champion), trained on equipment telemetry. risk_score is the calibrated probability (0-1) of an UNPLANNED failure within the next 30 days. 0.6 or more = act now (book preventive maintenance); 0.3-0.6 = monitor.
- **Revenue at risk:** launch_value estimates the revenue a launch delay would cost: weekly planned units x net price x weeks of delay avoided. All inputs live in value_assumptions and are ESTIMATES that the business can edit.
- **Action brief:** launch_action_brief contains an AI-generated (ai_query) summary of why a launch is at risk and what to do, in English and Italian. It is advice, not a decision.
- **Actions:** Planners escalate blockers in the Control Tower app; actions are stored in Lakebase and copied to ops_launch_actions. An open action has an owner and a due date.

9. Example questions the agents handle

Demand & Supply

- How is forecast accuracy trending by therapeutic area, and where is bias worst?
- Which launches are at risk, and is it master data, serialization, or artwork blocking them?
- Which SKUs have the highest expiry-at-risk and low OTIF?
- Quali lanci sono a rischio e cosa li blocca? (IT)

Manufacturing & Reliability

- What is our OEE by plant, with availability, performance and quality?
- What are the top root causes of major deviations?
- Which critical equipment is predicted to fail in the next 30 days, and what production is at risk?
- Qual e l'OEE per stabilimento? (IT)